

1. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        int out = open("output.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

        int err = open("error.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

        if (out < 0 || err < 0) {
            perror("open");
            exit(1);
        }

        dup2(out, STDOUT_FILENO);
        dup2(err, STDERR_FILENO);

        close(out);
        close(err);

        execlp("ls", "ls", "nonexistent_file", NULL);

        perror("ls");
        exit(1);
    }

    wait(NULL);
}
```

```
wait(NOERR);

printf("Output written to output.txt\n");
printf("Error written to error.txt\n");

return 0;
}
```

```
bindu@Bindusree:~$ nano skill14.c
bindu@Bindusree:~$ gcc skill14.c -o skill14
bindu@Bindusree:~$ ./skill14
Output written to output.txt
Error written to error.txt
bindu@Bindusree:~$
```